Assignment: Problem Definition in Human-Centered Design

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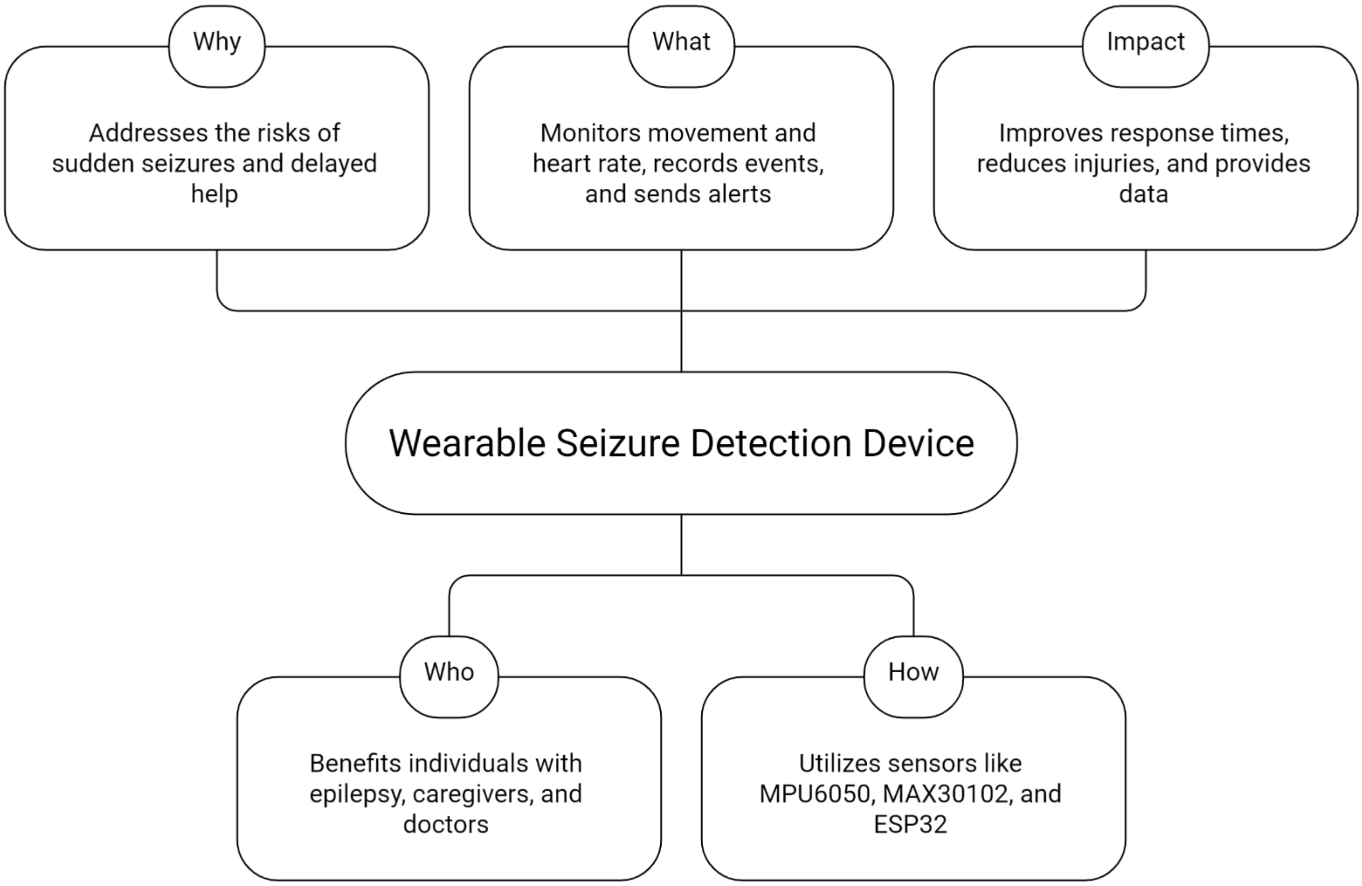
Problem Statement

Wearable Tonic-Clonic Seizure Detection and Alert System

Predict. Prevent. Protect.

Epilepsy is a neurological disorder in which a person experiences **repeated seizures** due to abnormal electrical activity in the brain. During some seizures, especially **tonic-clonic seizures**, the person may lose control of body movements and fall suddenly. This can lead to injuries or delayed medical help. Wearable technology can help by **detecting abnormal movements, monitoring vital signals, and sending alerts to caregivers** in real time.

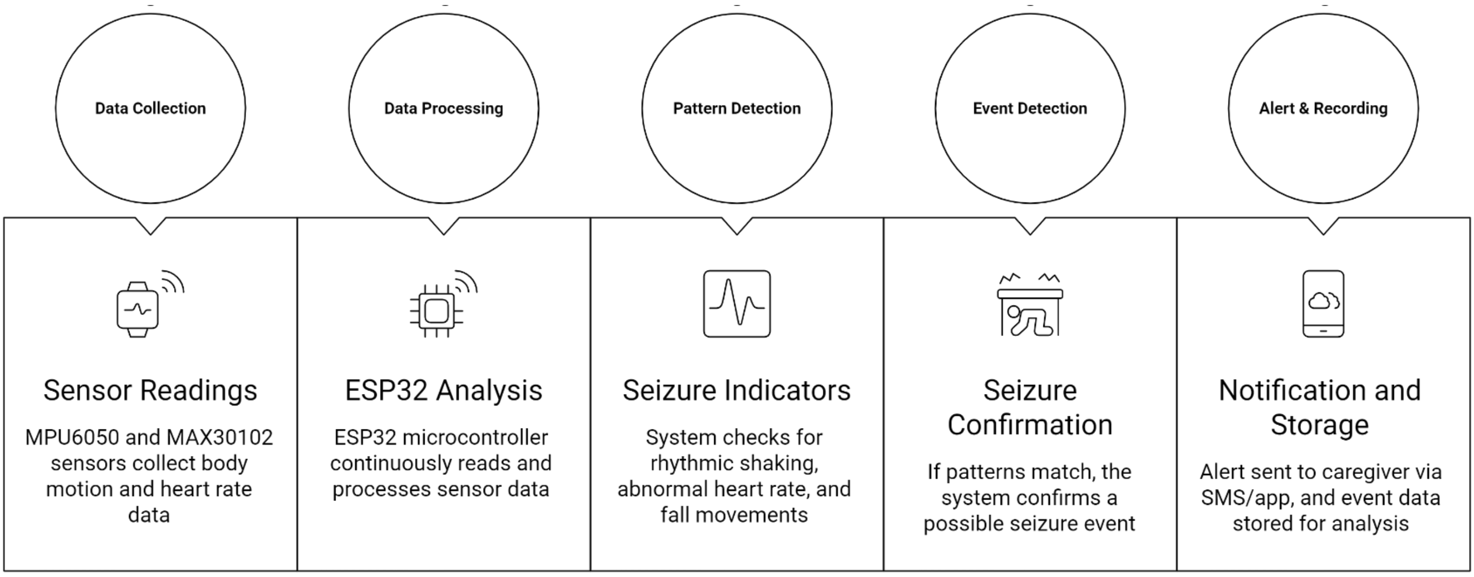
Why || Who || What || How || Impact



Community / User Group

People Living with Epilepsy – Individuals who experience seizures and need continuous monitoring and safety support.

Technical Flow



Stakeholder

Stakeholder	Role	Relevance / Interest
Epilepsy Patient	Primary User	Needs continuous monitoring, early seizure detection, and personal safety
Caregiver / Family Member	Primary Support User	Receives alerts and provides immediate assistance during seizures
Doctor / Neurologist	Secondary User	Analyzes recorded seizure data for diagnosis and treatment planning
Hospital / Healthcare Provider	Secondary Stakeholder	Supports patient monitoring and emergency response
Emergency Medical Services	Tertiary User	Responds quickly during severe seizure emergencies
Device Developer / Engineer	Technical Stakeholder	Designs, develops, and improves the wearable system
Researcher	Supporting Stakeholder	Uses data to improve seizure detection methods
Health Institutions	Regulatory / Advisory Stakeholder	Ensures safety, compliance, and ethical use

Problem Framing & Ideation

I noticed that people suffering from epilepsy often face uncertainty and fear because seizures can occur suddenly without warning. In some cases, patients live alone or are asleep when a seizure happens, which increases the risk of injury and delayed medical help. One of my classmates shared a story about a family member with epilepsy who experienced frequent seizures and needed constant supervision. I spoke to others and realized that many epilepsy patients struggle with the same issue—there is no simple way to continuously monitor them or alert caregivers immediately during a seizure. Using “How Might We” questions, I explored the problem further:

How might we detect seizures early using wearable technology?

How might we alert caregivers quickly without relying on constant human supervision?

I studied existing solutions such as hospital-based monitoring systems and mobile apps and found that they are either expensive, or some devices asking for subscription, inconvenient for daily use. This led me to the idea of a wearable seizure detection and alert system that can be worn on the wrist. By using motion and physiological sensors, the device can detect abnormal patterns and automatically notify caregivers.

Thus, I framed the problem as: people prone to seizures need a simple, affordable, and continuous monitoring system that can detect seizures early and provide timely alerts to improve safety and quality of life.

Impact Assessment

Evaluate user satisfaction and usability by wearing the fit band